



Managing heat rejected from air conditioning systems to save energy and improve the microclimates of residential buildings

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ABSTRACT

Simulations are conducted to investigate the influence and improvement potential of air conditioning heat rejection management of residential buildings on microclimate and energy use. The microclimate and building energy use are simulated on a typical high-rise building in Taipei, Taiwan, on a summer night. Heat rejection from the air conditioners is estimated with a building energy program, EnergyPlus, and a computational fluid dynamics (CFD) program, Windperfect, is used to analyze how heat rejection affects the outside thermal environment. Results show that heat rejection from air conditioners worsens the thermal environment below the urban canopy, thus increasing building energy use. Three counter-measure cases of heat rejection management, which consider the type of cooling system and its installation position, are proposed in this study. The average air temperature increase around the buildings caused by heat rejection was analyzed by transferring the simulation results of the building energy program to the CFD model on an hourly basis. Results show that the air temperature next to the building envelope and the air temperature around air conditioners decrease and that there is a reduction in electricity consumption by the air conditioners when a split-type air conditioner is installed on each floor or on every third floor. A reduction in the ambient air temperature below the urban canopy can be obtained by placing a cooling tower on the roof of the building.

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1. Introduction

The thermal properties of urban surfaces, the generation of anthropogenic heat, and the lack of evapotranspiration result in a significant temperature difference between urban and rural areas. The high-temperature urban areas are known as urban heat islands (UHIs). UHIs worsen the microclimate and increase air pollution, contributing to higher energy bills and greater human discomfort (Aikawa, Hiraki, & Eiho, 2006; Kolokotroni, Zhang, & Watkins, 2007; Solecki et al., 2005).

Buildings' energy use is affected by local conditions within the urban canopy, which is the space occupied by the buildings up to their roofs (Oke, 1976). A higher temperature in the city center increases the cooling load of buildings. The widespread use of air conditioning in urban areas has resulted in a significant increase in electricity consumption. The relationship between energy consumption and air temperature was investigated on a city-block-scale in Tokyo; the ratio of electricity consumption to air temperature during summer was 1.1–1.4 W/(floor-m²)/°C on

average (Ihara, Genchi, Sato, Yamaguchi, & Endo, 2008; Ihara, Kikegawa, Asahi, Genchi, & Kondo, 2008).

In order to evaluate the demand for cooling energy, individual air conditioning use schedules must be investigated. The heat rejected by air conditioning systems within the urban canopy structure was estimated using the software AUSSSM TOOL by Tanimoto, Hagishima, and Chimklai (2004), who found that the total anthropogenic heat released from HVAC systems varied over time. Peak cooling demand in residential units was estimated based on the behavior of occupants (Tanimoto, Hagishima, & Sagara, 2008); the research methodology accurately estimated the maximum cooling requirement for individual dwellings using probabilistic variations in occupant behavior. The spatial distribution and diurnal variation of the amount of heat rejected from air conditioning were obtained in order to evaluate the outdoor microclimate at night in residential areas of Taipei where air conditioners were installed in buildings. It was found that the peak of cooling demand in residential areas of Taipei occurred at night (Hsieh, Aramaki, & Hanaki, 2007a, 2007b).

The anthropogenic heat discharged from air conditioners has an impact on the outdoor thermal environment. For instance, Bojic, Lee, and Yik (2002) used simulations to demonstrate a significant increase in outdoor air temperature caused by the heat expelled

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from air conditioners in their investigation of air temperature and flow fields inside a recessed space exterior in high-rise buildings in Hong Kong. Fan and Sailor (2005) showed that anthropogenic heating plays an important role in the formation of urban heat islands by incorporating anthropogenic heating profiles into the planetary boundary layer (PBL) scheme in the MM5 meso-scale atmospheric model in a Philadelphia case study. Hsieh et al. (2007a, 2007b) conducted a simulation of a residential area in Taipei and observed that once outside air becomes hotter from the heat discharged from cooling systems, air conditioning systems consume more energy for cooling processes. The simulation results showed that higher temperatures were found around buildings, which increased the building cooling demand. Although mitigation strategies were not mentioned, these collective studies clearly demonstrate a relationship between the usage of air conditioning and the thermal environment in urban areas.

Recently, countermeasures proposed to improve the thermal environment and reduce buildings' energy needs have been discussed. Mitigation strategies such as the direct and indirect effects of cool roofs and shade trees around buildings were evaluated using the DOE-2.1E model in the US (Akbari & Konopacki, 2005). The application of distributed energy systems, which provide power and heat, in the dense Sancha area of Tokyo was examined in terms of energy consumption reduction (Chen et al., 2008). The centralized disposal of waste heat from the distributed local energy supply system had virtually no thermal influence on the heat island phenomenon within the area of analysis because the heat rose high into the upper atmosphere. The management of exhaust heat from cooling systems from the meso-scale point of view has been suggested. However, low condensers, which reject waste heat near street height, cause people discomfort and increase building energy use in the summer. Placing some or all of the condensers on top of the buildings and discharging the waste heat high above the buildings' roofs has been shown to have less of an impact on air temperature below the urban canopy (Dhakal & Hanaki, 2002). The impact of anthropogenic heat distribution on cooling energy demand was analyzed using the MM-CM-BEM simulation system set on a meso-scale meteorological model by Kikegawa, Genchi, Yoshikado, and Kondo (2003), using both the one-dimensional urban canopy model and building energy analysis model. Anthropogenic heat discharged near the ground increased the air temperature by 0.62 °C, and increased daily cooling energy demand by 2.79% compared to anthropogenic heat discharged from the rooftop. Near-ground summer temperatures were found to decrease after individual countermeasures were implemented in residential and office canopies, respectively (Kikegawa, Genchi, Kondo, & Hanaki, 2006). For office buildings in Tokyo, the reduction in anthropogenic heat from air conditioning was shown to effectively reduce cooling energy on a city-block scale (Ihara, Genchi et al., 2008; Ihara, Kikegawa et al., 2008). The above studies investigate the thermal environment and energy use from a large scale viewpoint. The details of vertical and horizontal temperature distributions and the effect of air temperature next to the building envelope on building energy use were not discussed.

The measured heat island intensity, the biggest temperature difference between the center of Taipei City and its suburban areas, was 4.9, 4.5, and 4.5 °C, respectively, at 14:00, 21:00, and 02:00 in the summer (Lee, 1999). Heat rejection, which is regarded as one of the main causes of heat islands in residential areas, increases the temperature within the urban canopy and thus increases the electricity consumption of buildings. However, there are few options for air conditioning systems in residential buildings that influence the microclimate and building energy use. The objective of the present study is to clarify how the heat rejection management affects the thermal environment and building energy use on the micro-scale in residential areas during summer nights in Taipei

City. A computational fluid dynamics (CFD) model and a building energy program are used to simulate the effects of type of cooling system and installation position on outside air temperature and energy use. In the CFD model, each building is constructed to scale and the elevation and position of the air conditioning systems are shown to provide details about the temperature distribution caused by the exhaust heat from cooling systems below the urban canopy. Furthermore, the lifestyle of people living in Taipei and the percentage of people who use air conditioners are taken into account in the calculations of the buildings' hourly heat rejection.

The rest of this paper is organized as follows. Section 2 introduces the research methodology, including the CFD model and the building energy program. Terms related to the simulation are also defined. The three test cases are defined and the microclimate simulation results are given in Section 3. Power consumption is discussed in Section 4, and conclusions are given in Section 5.

2. Research methodology

Mitigation scenarios from the viewpoint of heat rejection management (cooling system types and installation positions) are discussed in this paper to improve the thermal environment at night by using a building energy program and a CFD model. The building energy program estimates the amount of heat rejection on an hourly basis according to a time schedule of air conditioner use throughout the night. For the CFD model, the building geometry was replicated using a geographic information system (GIS) with a resolution of more than three million meshes. The initial condition in this paper is described as the simulation results obtained using the original weather data, with the influence of heat rejection not considered during the simulations. The CFD simulation, running in continuity, is provided with hourly data of weather and heat rejection discharged from air conditioners. The original weather data is revised using the average increase in temperature in the focus area obtained from the CFD simulation. The revised weather data is then used to estimate the additional electricity consumption of air conditioners using EnergyPlus. The microclimate within the urban canopy for a given hour is influenced by that of the previous hour. The CFD simulation was conducted from 18:01 to create the initial condition for 19:01. The increased temperature around the buildings due to heat rejection by air conditioners was analyzed for Cases 0–3 using the CFD model, and the additional electricity consumption of air conditioners caused by the increase in air temperature was estimated using the building energy program.

Three cases of heat rejection management that consider various cooling system types and installation positions are examined in this study. Case 1 includes a split-type air conditioner installed on each floor, where 2–3 rooms share an exterior unit. The number of exterior units is therefore reduced from 16 per floor, as in the current situation in the study area, to 8. Case 2 includes a split-type air conditioner for one family for which the elevation of heat rejection is increased and where spaces are designated on every third floor for exterior units. The waste heat from floors $3n-2$ and $3n-1$ is discharged on floor $3n$, where n is a positive integer. Finally, in Case 3, a cooling tower, which primarily uses latent heat from vaporization (evaporation) to cool processed water, is installed. The air provides minor additional cooling due to the temperature increase (sensible heat). In order to focus on the heat rejection management of the cooling systems, the coefficient of performance (COP) of the air conditioners in the three cases is specified to be the same as that of the window-type air conditioners described in a previous study (Hsieh et al., 2007a, 2007b).

In order to perform an integrated analysis of building energy use and the thermal environment, a building energy program

(EnergyPlus) and CFD software (Windperfect) are used in this study. EnergyPlus is used to evaluate the energy performance of buildings, including the estimation of heat rejection and electricity consumption, whereas Windperfect, which is designed for airflow modeling, is used to determine the hourly distribution of air temperature, humidity, and wind speed. Based on structured grids, Windperfect calculates airflow using the Cartesian finite volume method with a well-designed interface for pre- and post-processing.

The simulation conditions for the building materials and window-wall ratio are set according to Taiwan's Building Energy Conservation Design Calculation Example (Construction Magazine Publishing House, 2002). Data on the summertime use of air conditioners in Taipei from a report by the Taiwan Power Company (2002) were used to calculate the hourly heat rejection of the buildings. The air conditioners in this study are set at 25 °C.

The study area for the simulation is the Cheng Kung public housing complex, typical of those in Taiwan, consisting of a group of high-rise buildings with a high population density. The complex consists of 38 buildings, 32 of which are more than 15 stories tall with an average floor area per story of about 424.3 m². Additionally, many of the units in the buildings have window-type air conditioners. Fig. 1a shows the building geometry and focus area (the area covered by the finest mesh) viewed from the southeast. The building geometry is created using GIS data, including land use and building information. A total of 4-level meshes is in the simulation. The entire simulation boundary was 1200 m * 1200 m * 400 m in the x, y, and z axes, respectively. The highest building in the simulation domain was a 25-floor apartment building ($H = 75$ m), which is located on the east side of the focus area. The computational domain was constructed with a vertical length of 5H, and lateral and upwind lengths of more than 3H.

The size of the finest mesh, located at the center of simulation area, was 2 m * 2 m in the x and y axes, respectively. For the surrounding area, the mesh size gradually became coarser (in increments of 3, 5, and 10 m) with distance from the center to the outside. The vertical axis was set at 1 m in the focus area, and gradually increased with height. The total number of meshes in the CFD simulation was over three million. Wind velocity as a function of height above the ground was assumed to follow a power law distribution. The exponent was set to 0.27, which is the value in the city area. The meteorological data is listed in Table 1.

EnergyPlus was used to measure the heat rejection of buildings, which was based on an energy use simulation of residential buildings according to residential air conditioner use in Taipei. The percentages of people who turned on their air conditioner were 32.90% for the period 18:01–21:00, 75.8% for the period 21:01–24:00, and 47.9% for the period 00:01–06:00. The temperature distribution outside the buildings was then analyzed using Windperfect. In the simulation, the previously calculated heat rejection was input into Windperfect to evaluate how the heat discharged from buildings affects the outside thermal environment. After evaluating the temperature distribution around buildings below the

Table 1

Meteorological data simulation settings (location: 25°04'N in latitude and 121°33'E in longitude, elevation: 6 m above sea level).

	Air temp. (°C)	Wind speed (m/s)	Wind direction (°)
19:01–20:00 h	30.88	4.02	286.7
20:01–21:00 h	30.33	3.02	264.2
21:01–22:00 h	30.00	1.67	265.8
22:01–23:00 h	29.59	2.81	264.2
23:01–24:00 h	29.13	2.93	283.3
00:01–01:00 h	29.00	2.10	300.0
01:01–02:00 h	28.71	2.68	300.0

urban canopy, three temperatures (T_{am} , T_{bu} , and T_{ac}) were evaluated to clarify the thermal environment around the buildings for each case. Finally, the additional electricity consumed by air conditioners due to heat rejection from condensers was predicted using EnergyPlus.

It should be noted that all the temperatures were obtained using meshes of 2 m * 2 m * 1 m in the x, y, and z axes, respectively. The three temperatures used in the study are explained in detail below, followed by a description of hourly air temperature variations and spatial distribution during the time period 19:01–02:00.

(a) Ambient air temperature: T_{am}

In order to assess the temperature in the building canopy, the focus area was defined to calculate the ambient air temperature. The focus area, which is the space inside the frame in Fig. 1b, is located in the center of the building area; it contains six high-rise (19-floor) buildings. The area covered by the finest mesh (2 m * 2 m * 1 m), which amounts to 88 m * 144 m * 60 m, is defined as the focus area. Among the 197,100 meshes in the focus area, buildings occupied 53,412 meshes, with the remaining 143,663 meshes containing the atmosphere. T_{am} , the ambient air temperature, is defined as the average air temperature in the focus area.

(b) Air temperature around air conditioners: T_{ac}

A typical window-type air conditioner takes in air from its two sides and discharges heat to the outside, in a direction perpendicular to the building walls. T_{ac} is the average temperature in the meshes on the sides of air conditioning systems. It should be noted that the mesh here does not include the airflow outlet of the air conditioners. T_{ac} indicates the temperature of the air that enters the air conditioners.

(c) Air temperature next to building envelope: T_{bu}

T_{bu} is the average temperature in the meshes attached to the six buildings, i.e., the air temperature next to the building envelope. All four sides of the buildings and roofs, composed of 26,163 meshes, are considered. The temperature next to the buildings

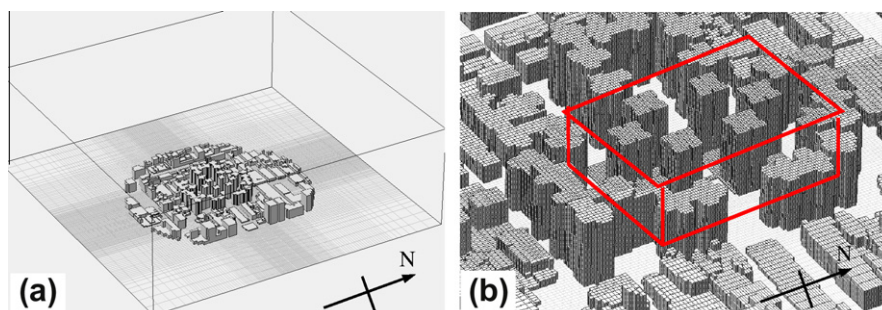


Fig. 1. Geometry (a) and the focus area (b) of CFD simulation.

dominated their heat balance; T_{bu} was thus input into EnergyPlus to calculate building energy use.

2.1. Coefficient of performance (COP) of cooling systems

The default cooling system in WindowAC.idf, an example file in EnergyPlus, is used. Five curves specify the DX coil performance. They include the total cooling capacity modifier curve (function of temperature), the energy input ratio (EIR) modifier curve (function of temperature), the total cooling capacity modifier curve (function of flow fraction), the energy input ratio modifier curve (function of flow fraction), and the part load fraction correlation (function of part load ratio).

The total cooling capacity modifier curve and the energy input ratio modifier curve are biquadratic curves with two independent variables (EnergyPlus, 2003): wet-bulb temperature of the air entering the cooling coil and dry-bulb temperature of the air entering the air-cooled condenser coil. The output of the curve is multiplied by the rated total cooling capacity to give the total cooling capacity at specific temperatures (i.e., at temperatures different from the rating point temperatures). The output of the curve is multiplied by the rated EIR (inverse of the rated COP) to give the EIR at specific temperatures. Both performance curves are normalized to have a value of 1.0 at the rating point temperature.

Meteorology data (wet-bulb temperature and dry-bulb temperature) and T_{ac} , the air temperature around air conditioners, were used as independent variables to obtain the performance curves. The increase of T_{ac} compared to the meteorology data ranged from 0.64 to 1.56 °C for the time period 19:01–02:00 (Hsieh et al., 2007a, 2007b). The performance curves of cooling capacity (function of temperature) and EIR (function of temperature) are shown in Fig. 2. The cooling capacity slightly decreased and the EIR slightly increased with increasing outside air temperature. Less than 3% of the average cooling capacity and EIR were observed at higher temperatures according to the performance curves. The influence of the COP due to the temperature increase resulting from waste heat rejection was neglected in this study in order to focus on heat rejection management and the height and position of cooling systems. Although all air conditioners in the countermeasure cases had the same COP value (2.90), in the building energy simulations, the feedback effects that the air temperature rejected from the condensers increased due to T_{ac} were concerned during the CFD simulation.

2.2. Heat rejection by sensible and latent heating

2.2.1. Sensible heating—split-type air conditioners

Split-type air conditioners take in air near the condenser and expel heat to the outside using a fan. The heat of the exhaust air

depends on indoor energy use and the air entering the air conditioner unit. Heat rejection from air conditioners is a process of sensible heating; that is, heat is added to the air without changing its relative humidity. The sensible heating process is represented on the psychrometric chart by straight horizontal lines. The humidity ratio of the air remains constant because no moisture is added or removed. The governing equations are:

$$Q_S = M * (H_B - H_A) = M * S * (T_B - T_A) \quad (1)$$

where Q_S sensible heat that causes an increase in dry-bulb temperature, H_i enthalpy of point i , kJ kg^{-1} , T_i temperature of point i , °C, M air flow rate, kg s^{-1} , S specific heat of the air, $1.005 \text{ kJ kg}^{-1} \text{ K}^{-1}$

2.2.2. Sensible and latent heating—the cooling tower

A cooling tower discharges heat in both sensible and latent forms. Mechanical draft cooling towers that use large fans to force air through circulated water are the most commonly used. The average relative humidity of the outlet airflow from this type of cooling tower is 92–96% (Stabat & Marchio, 2004). The water falls downward over fill surfaces, which increases the contact time between the water and the air.

Heating and humidification increase the dry-bulb temperature and humidity ratio of the air, respectively. The heat gain from the initial point to the final condition can be broken into sensible and latent heat portions. When removing heat from an object results in a temperature drop, the heat removed is called sensible heat, which is shown as a horizontal movement on a psychrometric chart. In contrast, heat absorbed or released as a result of a phase change without a temperature change is called latent heat, which is shown as a vertical movement on a psychrometric chart. Enthalpy is the heat content (energy) of the unit mass of the atmosphere. The intersection of a horizontal line through A (inflow air) and a vertical line through C (outflow air) occurs at D. The change in heat of a cooling tower can be expressed as:

$$\begin{aligned} Q_W &= Q_S + Q_L = G * (H_D - H_A) + G * (H_C - H_D) \\ &= G * (H_C - H_A) \end{aligned} \quad (2)$$

where Q_L latent heat, the heat required to evaporate a given amount of moisture, G air loading kg s^{-1} .

EnergyPlus was used to calculate the total amount of heat rejection. Air loading depends on cooling tower specifications. The humidity ratio is constant with regard to the horizontal movement (sensible heat) and the dry-bulb temperature is constant with regard to the vertical movement (latent heat). The temperature of the outlet air was evaluated based on the relative humidity of the outflow air, and the ratio of sensible heat to latent heat was calculated.

3. Microclimate simulation results

3.1. Case 0: current situation in study area

Fig. 3 shows the increases of T_{am} , T_{bu} , and T_{ac} , in Case 0, the current situation in the study area, from 19:01 to 02:00 and Fig. 4 shows the ambient temperature (T_{am}) by floor for Case 0, as described in a previous study (Hsieh et al., 2007a, 2007b). The x axis indicates the time period and the y axis indicates the temperature increase compared to the original weather data. T_{am} , T_{bu} , and T_{ac} in Case 0 are used as the reference temperatures to explain the effects of the cases in the following sections.

The countermeasure case of a waste heat reduction rate of 100% was evaluated using a multi-scale numerical simulation system, MM-CM-BEM. The simulation results reveal that the influence of waste heat on air temperature at a height of 3 m for low-sky-view-factor residential canopies was about 0.5 °C (Kikegawa et al.,

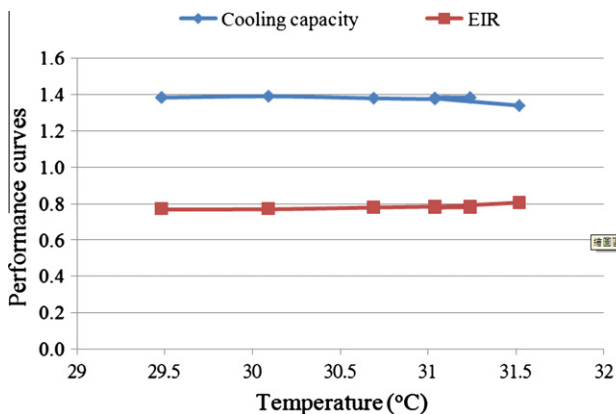


Fig. 2. The performance curves of the cooling system.

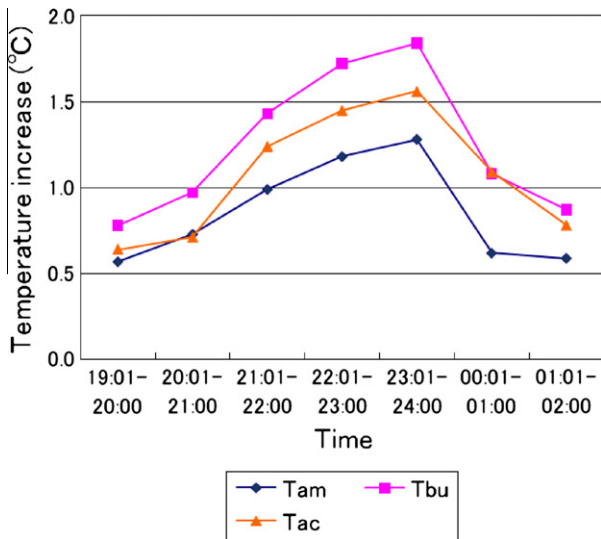


Fig. 3. The three specified temperature increases by hourly time period in Case 0.

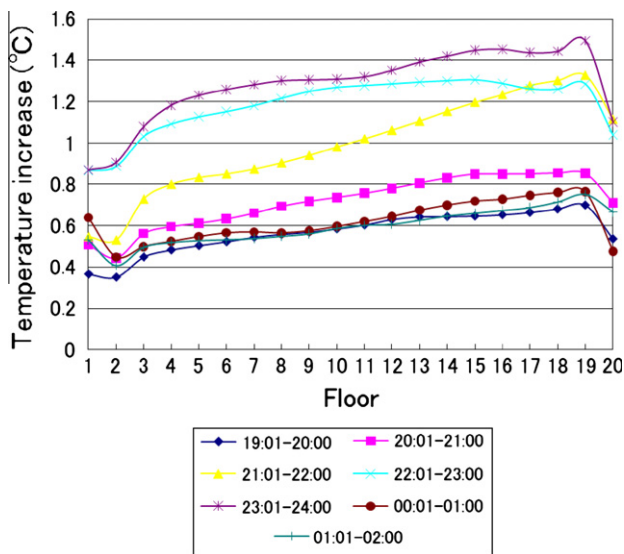


Fig. 4. The increase of the ambient temperature (T_{am}) by floor in Case 0.

2006). The sky view factor is the ratio of sky areas when looking up from the ground. The average ambient temperature increases in the present study at heights of the 1st, 2nd, and 3rd floors were 0.61 °C, 0.56 °C, and 0.70 °C, respectively. Similar results of air temperature near the ground surface were obtained.

3.2. Case 1: split-type I air conditioner

Fig. 5 shows the temperature distribution in the mesh next to the building surfaces in Case 1 as viewed from the southeast. The bold line shows the boundary of a 1.0 °C temperature increase from the original weather data. The air temperatures, indicated by a gradient from white to black, are in a range of 28–33 °C. The higher temperatures next to the buildings indicate the positions of exterior units. There are two exterior units on each side of the buildings in the focus area. Fig. 6 shows increases in T_{am} , T_{bu} , and T_{ac} , in Case 1 compared to those in Case 0. The ventilation around the buildings was improved by the decrease in the number of exterior units; decreases in T_{bu} and T_{ac} , as compared to those in Case 0, were thus observed. The decrease of T_{bu} indicates a slight reduction

in building energy use and the decrease of T_{ac} indicates a temperature reduction of the outlet airflow of the air conditioners. Nevertheless, a stagnation of heat below the urban canopy was still observed because the waste heat was discharged on every floor. Therefore, T_{am} , the average ambient temperature in the focus area, exhibited no obvious change compared to that in Case 0. Fig. 7 shows the ambient temperature (T_{am}) increase by floor in Case 1 as compared to the original weather data. The waste heat discharged from air conditioners moved upwards because higher temperature air has a lower density. It was observed that the ambient air temperature gradually increased with height. The average ambient air temperature difference between the 1st and 19th floors was 0.35 °C from 19:01 to 02:00. However, the average maximum temperature difference between floors from 19:01 to 02:00 was observed between the 2nd and 17th floors. The 1st floor is a hallway for pedestrians where the outside air can only flow through the entrance and exit of the hallway, giving it poor ventilation. Better ventilation is found on the 2nd floor, which is an open space. The waste heat on top of the building was carried away by wind, so the maximum temperature increase was found at the 17th floor.

3.3. Case 2: split-type II air conditioner

Fig. 8 shows the temperature distribution in the mesh next to the building surfaces in Case 2 viewed from the southeast. The meshes near the exterior units, installed on every third floor and the tops of the buildings, showed higher air temperatures. Fig. 9 shows the increases in T_{am} , T_{bu} , and T_{ac} in Case 2 as compared to those in Case 0. The simulation results show that the three temperatures in Case 2 are lower than those in Case 0. Exterior units installed on every third floor decreased the temperature around the building (temperatures T_{bu} and T_{ac}). Compared to Case 0, the decrease in T_{bu} indicates lower building energy use and the decrease in T_{ac} indicates a reduction in the temperature of the air conditioners' outlet airflows. The portion of waste heat discharged from the roof of the buildings (the waste heat from the 16th–19th floors) thus improved the thermal environment in the urban canopy. The average T_{am} during the time period 19:01–02:00 in Case 2 was 0.28 °C lower than that in Case 0. Fig. 10 shows the ambient temperature (T_{am}) by floor for Case 2 compared to the original weather data. The exterior units were located on the 3rd, 6th, 9th, 12th, and 15th floors, and the roof. Therefore, the air temperatures at these elevations were higher. No waste heat was discharged near the ground surface, and therefore a large decrease in temperature was observed on the 1st floor compared to Cases 0 and 1. The waste heat from 16th–19th floors discharged from the top of the buildings increased the elevation of heat rejection, so the observed ambient temperature (T_{am}) increase below the urban canopy was smaller than those of Cases 0 and 1. The average maximum temperature difference between floors from 19:01 to 02:00 was observed between the 1st and 15th floors.

The countermeasure case of a waste heat reduction rate of 50% was evaluated using a multi-scale numerical simulation system, MM-CM-BEM. The simulation results reveal that the influence of waste heat on air temperature at a height of 3 m for low-sky-view-factor residential canopies was about 0.25 °C (Kikegawa et al., 2006). The average ambient temperature increases in the present study at heights of the 1st, 2nd, and 3rd floors were 0.13 °C, 0.24 °C, and 0.52 °C, respectively. Similar results of air temperature near the ground surface were obtained.

3.4. Case 3: cooling tower

A mechanical draft and indirect-contact evaporative cooling tower was used in this study. The estimations of the percentages

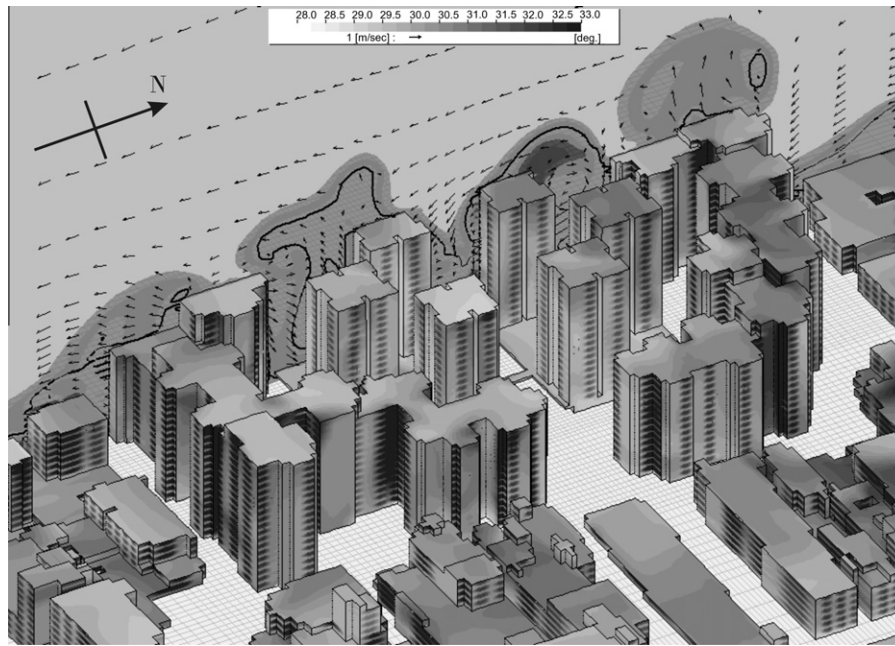


Fig. 5. Temperature distribution from a SE view of Case 1 from 23:01 to 24:00.

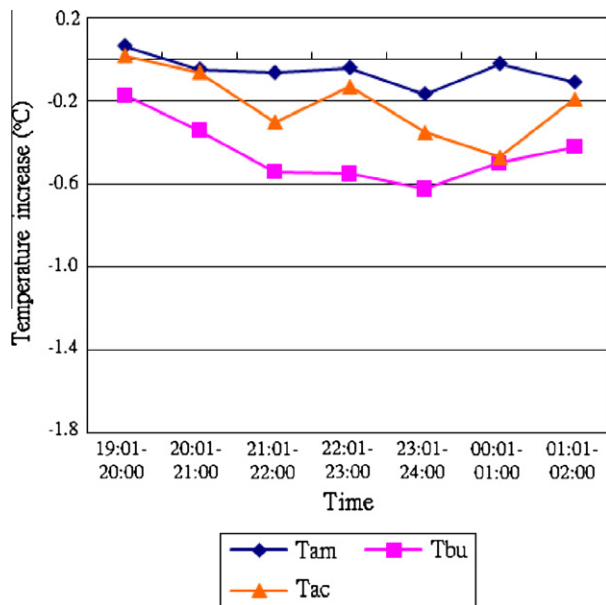


Fig. 6. The three specified temperature increases by hourly time period in Case 1.

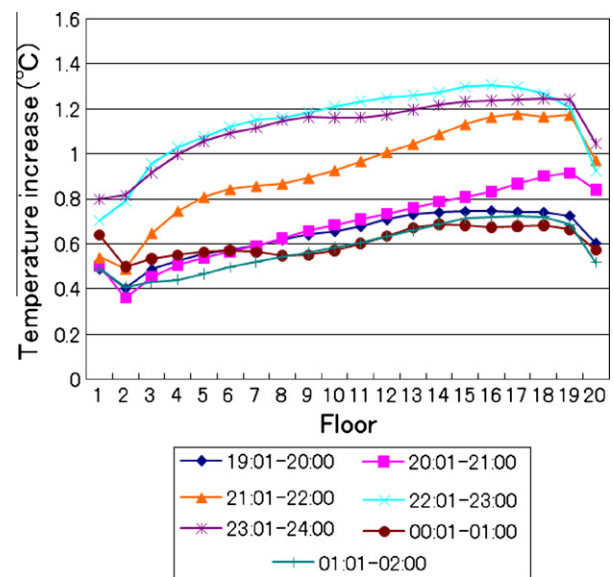


Fig. 7. The increase of the ambient temperature (T_{am}) by floor in Case 1.

of sensible and latent heat for each time interval were based on the assumption of 94% outlet airflow relative humidity. Fig. 11 shows the temperature distribution in the mesh next to the building surfaces in Case 3 viewed from the southeast. The sensible heat increased the air temperature around the cooling towers. The high-rise buildings outside the focus area were found to have higher temperatures due to heat rejection from nearby buildings that used window-type air conditioners. Fig. 12 shows the increases in T_{am} , T_{bu} , and T_{ac} in Case 3 as compared to those in Case 0. The cooling tower discharged waste heat on top of the buildings, and therefore significant decreases in all three temperatures were observed compared to Case 0. It was found that more than 80% of the heat was discharged in the form of latent heat at night. The maximum decrease in temperature was found during the time per-

iod of 23:01–24:00. Fig. 13 shows the ambient temperature (T_{am}) increase by floor for Case 3 compared to the original weather data. A large decrease in the temperature on the 1st floor was observed. The temperature increase at the top of the floor was smaller than those of Cases 0 and 1 because the waste heat was discharged in the form of latent heat.

The countermeasure case of buildings using cooling towers was evaluated using a meso-scale model. The simulation results reveal that the peak reduction in temperature was about 0.35–0.5 °C (Dhakal & Hanaki, 2002). The average ambient temperature reduction compared to Case 0 below the urban canopy was 0.57 °C in the present study. Similar mitigation effects of the cooling tower obtained from micro-scale simulations were confirmed with a meso-scale model. An obvious decrease in air temperature below

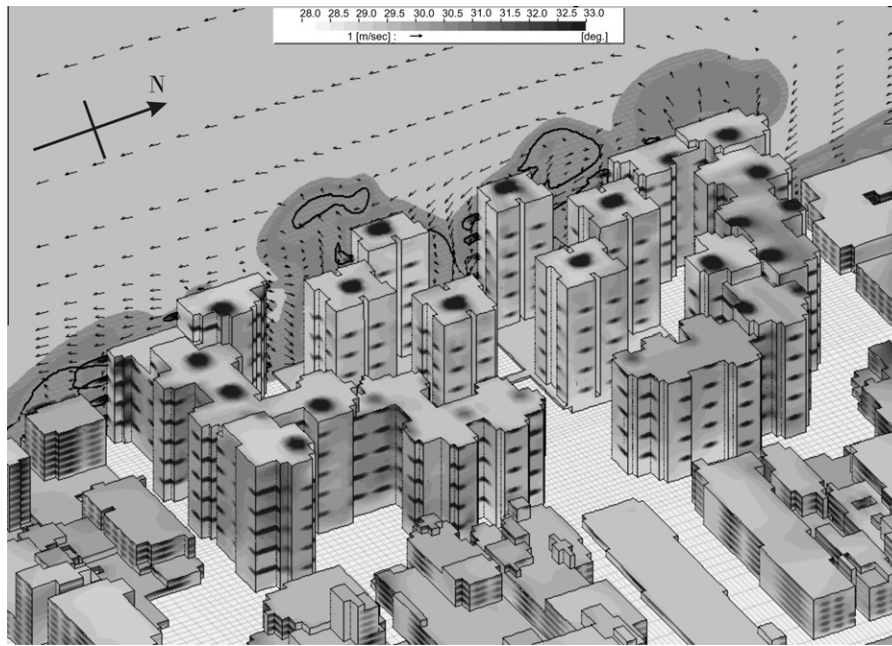


Fig. 8. Temperature distribution from a SE view of Case 2 from 23:01 to 24:00.

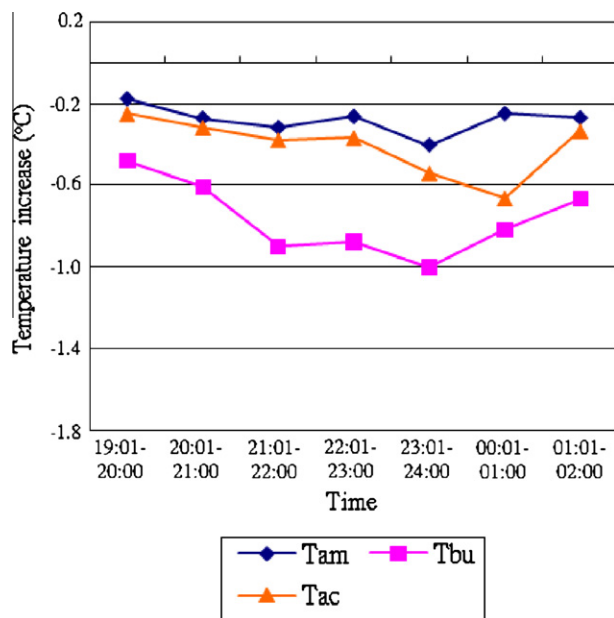


Fig. 9. The three specified temperature increases by hourly time period in Case 2.

the urban canopy was observed. The cooling tower thus efficiently improves the thermal environment below the urban canopy.

4. Electricity consumption

The air temperature next to the building envelope (T_{bu}) directly dominated the heat balance of the buildings as well as the electricity consumption of the air conditioners. Fig. 14 shows the temperature increase around the buildings (T_{bu}) for the three cases by floor as compared to Case 0 during the period 23:01–24:00, which was the time period of maximum cooling demand. T_{bu} was affected by the location of exterior units. T_{bu} values of Case 2 on the 3rd,

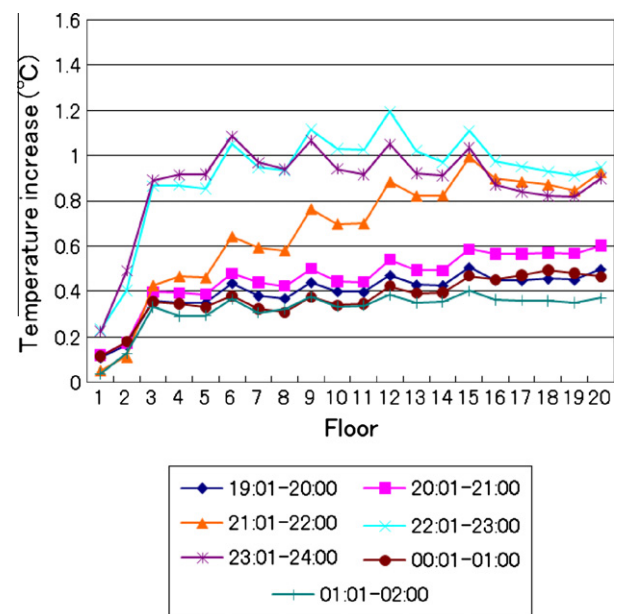


Fig. 10. The increase in ambient temperature (T_{am}) by floor in Case 2.

6th, 9th, and 12th floors were all higher than those of Case 1, and those of Case 3 were lower than those of Cases 1 and 2. An improvement of microclimate was observed for all three cases on most floors; the temperature reduction in Case 3 was most obvious. Fig. 15 shows the increases in air temperature around buildings for the three countermeasure cases compared to Case 0. Decreases in average temperature around the buildings (T_{bu}) for all three cases were found during the period 21:01–24:00. The improvement of the thermal environment (the reduction of T_{bu}) in Case 1 was the smallest, and that in Case 3 was the largest. The reduction of T_{bu} compared to Case 0 was obvious during the period 21:01–24:00, when 75.8% of residential areas in Taipei use air conditioners. The average reductions in Cases 1, 2, and 3

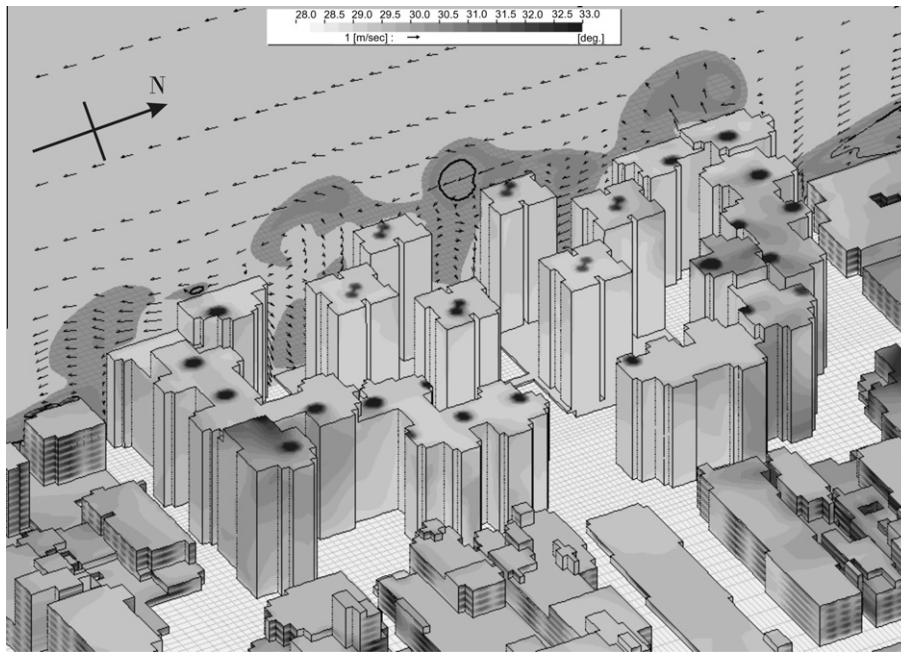


Fig. 11. Temperature distribution from a SE view of Case 3 from 23:01 to 24:00.

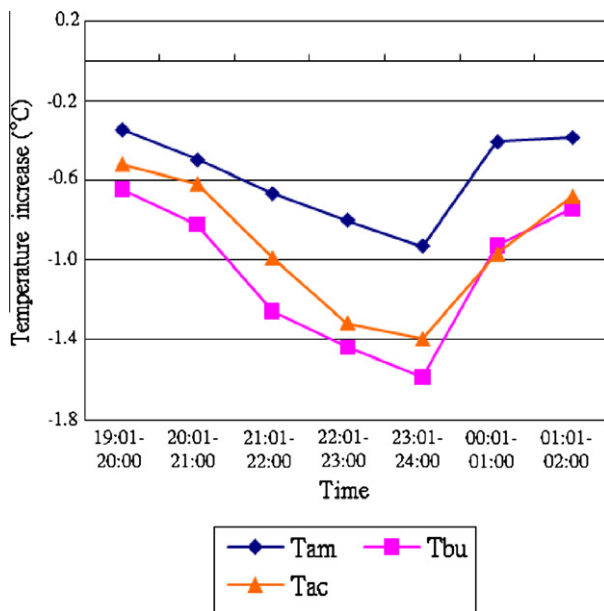


Fig. 12. The three specified temperature increases by building height in Case 3.

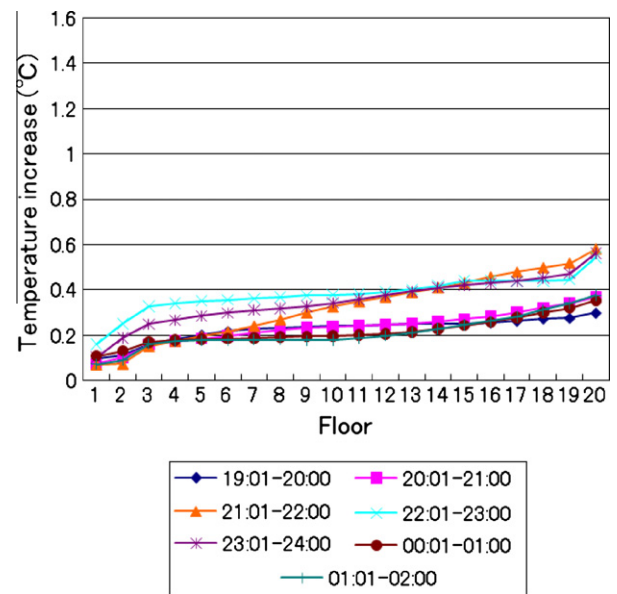


Fig. 13. The increase in ambient temperature (T_{am}) by floor in Case 3.

were 0.45, 0.76, and 1.06 °C from 19:01 to 02:00, respectively. Fig. 16 shows the electricity consumption per square meter of floor area at night for each case. The difference among each countermeasure indicates the potential of energy saving effect. The initial condition represents electricity consumption without considering the influence of heat rejection, as discussed in a previous study (Hsieh et al., 2007a, 2007b). Cases 0, 1, and 2 show electricity consumption with the influence of heat rejection taken into consideration. The air temperature becomes high due to the heat discharged from cooling systems, which in turn increases the energy consumption of the cooling systems. The feedback of heat rejection to air conditioning load is defined as the increase of electricity consumption that results from the heat rejection of air conditioners. The feed-

back of heat rejection to air conditioning loads in Cases 1 and 2 during the time period 19:01–02:00 were 4.26 W/m² and 2.73 W/m², respectively. The additional electricity consumption rates of the air conditioners in Case 1 were 0.72 W/m², 1.04 W/m², and 1.34 W/m² during each hour from 21:01 to 22:00, 22:01 to 23:00, and 23:01 to 24:00, respectively, or 7.2%, 9.7%, and 13.0% higher compared to Case 0, respectively. The additional electricity consumption rates of the air conditioners in Case 2 were 0.43 W/m², 0.67 W/m², and 0.96 W/m² during each hour from 21:01 to 22:00, 22:01 to 23:00, and 23:01 to 24:00, respectively, or 4.6%, 6.9%, and 8.0% higher compared to Case 0, respectively. Although additional electricity compared to that for Case 0 was required, energy savings were observed in Cases 1 and 2.

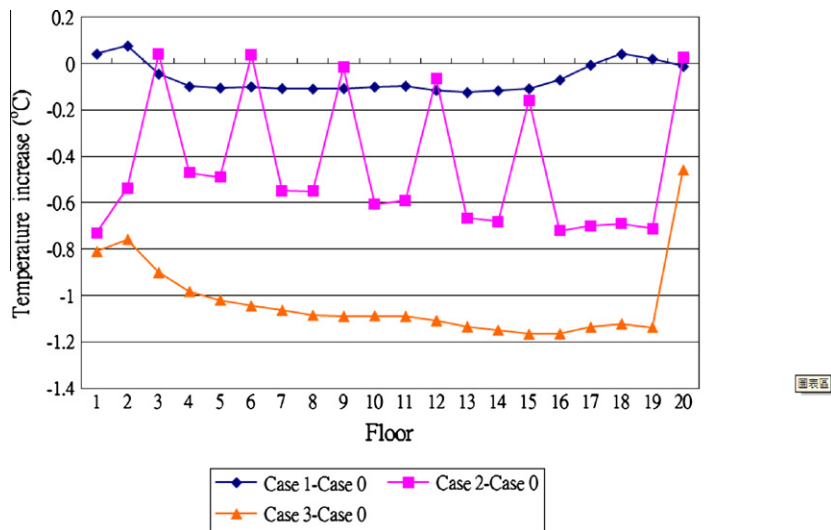


Fig. 14. The increase in air temperature around buildings (T_{bu}) from 23:01 to 24:00.

5. Conclusion

Anthropogenic heat discharge from air conditioners increases the temperature in the city center, which results in a significant increase in electricity consumption. Previous studies highlighted the crucial influence of anthropogenic heating on building energy use. This study, from the mitigation viewpoint, evaluated the potential of heat rejection management of cooling systems improving the microclimate and building energy use by using a micro-scale approach. A building energy program and CFD software were used with hourly weather information data and individual air conditioning use schedules. This combination (CFD software and building energy program) provides both spatial and temporal simulation results, allowing the temperature distribution and the effect of air temperature next to the building envelope on building energy use to be determined. A previous study (Hsieh et al., 2007a, 2007b) reported that for window-type air conditioners, higher temperature due to heat rejection was observed, which increased building cooling demand. In order to improve the microclimate and building energy use, mitigation scenarios for night-time heat rejection management for residential buildings in Taipei were proposed and simulated in this study. Three countermeasure cases of heat rejection management, which consider the type of cooling systems and their installation position, were analyzed. Simulation results were compared with those from studies that used meso-scale models (Dhakal & Hanaki, 2002) and a multi-scale numerical simulation system (Kikegawa et al., 2006). Furthermore, the vertical and horizontal temperature distributions, including the average ambient temperature, the temperature next to the building envelope, and the temperature around air conditioners within the urban canopy, were evaluated.

Simulation results show that the ambient temperature did not obviously decrease within the urban canopy when split-type air conditioners on each floor were used, indicating that the microclimate was not improved. However, building energy use was slightly reduced due to a decrease in temperature next to the buildings and around air conditioners, confirming the potential energy savings from using split-type air conditioners on each floor. Although there was no obvious difference in the heat exhaust from split-type air conditioners on each floor and on every third floor, the simulation results reveal that split-type air conditioners on every third floor, which increase the elevation of heat rejection, reduce the ambient temperature and the temperature around buildings. Feedback values of heat rejection to the air conditioning load, defined as

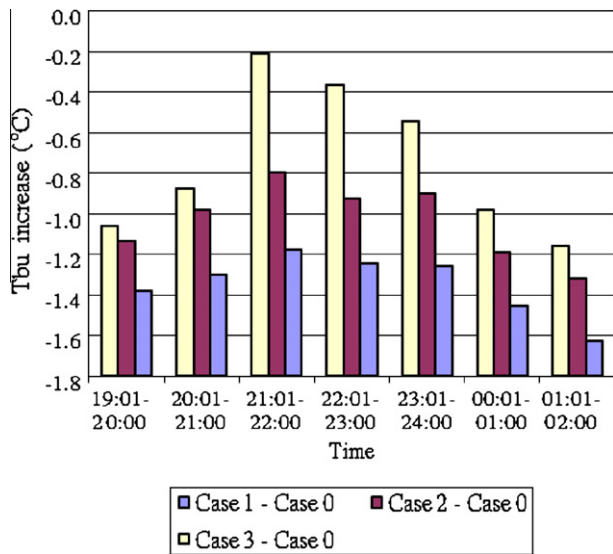


Fig. 15. The increase in air temperature around buildings (T_{bu}) in each case.

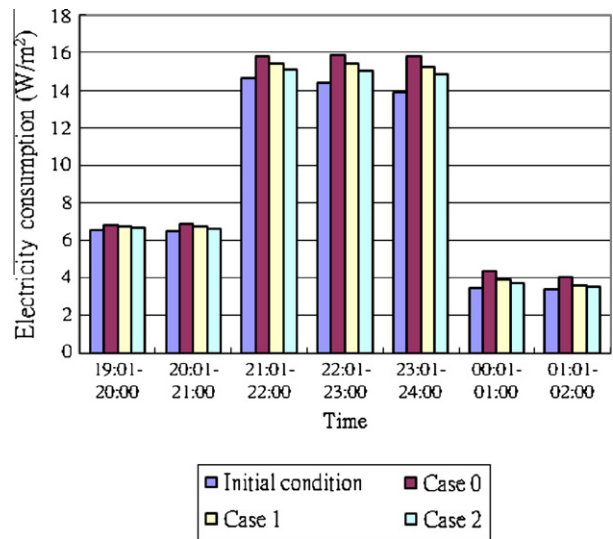


Fig. 16. Electricity consumption of air conditioning systems.

the increase of electricity consumption that results from the heat rejection of air conditioners, were introduced to discuss the building energy use. Simulations confirm that split-type air conditioners on every third floor reduce building energy use during summer nights and exhibit a better improvement for a given air conditioner schedule. The mechanism of heat rejection of a cooling tower was presented and the improvement in the thermal environment due to the discharge of heat in the form of latent heat was shown. Although high-rise buildings near the edge of the focus area had higher temperatures due to heat rejection coming from nearby buildings, an obvious reduction in air temperature below the urban canopy was observed when a cooling tower was placed on the roof of the building.

Results show that placing the exterior part of air conditioners, or installing cooling towers, on the roofs of buildings discharges heat high above the street, reducing its impact on the air temperature below the urban canopy. A follow-up study on the interaction between exhaust heat from air conditioners and the urban thermal environment using micro-scale methodology is planned. Moreover, the influence of the coefficient of performance for air conditioners due to the temperature increase from waste heat rejection on building energy use will be considered for future model improvement.

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